



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026-27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 10

Team No.: 14

Project Title: Linux File System Explorer Using System Calls

Team Members

Roll Number	Student Name	Signature
2520090033	V Vishwa Teja	
2520090058	B Sree Nikhil Reddy	
2520090191	Nalla Rohith	

1. Abstract

Argument passing is an important concept in programming that allows information to be transferred from one function to another. When a function is called, we can provide it with certain values, known as arguments, which the function uses to perform a specific task. This makes programs easier to understand, organize, and reuse. There are mainly two common ways of passing arguments: **pass by value** and **pass by reference**. In pass by value, a copy of the original value is given to the function. Any changes made to that value inside the function will not affect the original variable. On the other hand, pass by reference provides the function with access to the original variable or its reference. Therefore, changes made inside the function can also affect the original data. Argument passing is useful because it allows programmers to create flexible functions that can work with different inputs instead of writing separate code for every situation. It also helps reduce repeated code and makes programs more modular and manageable. Different programming languages handle argument passing in different ways. For example, C commonly uses pointers to achieve reference-like behavior, while Java passes primitive values by value and object references by value. Python also passes object references by assignment.

2. Problem Statement

In Linux-based systems, programs often need to receive input directly when they are executed. Manually entering information after the program starts can make simple tasks less convenient and less flexible. Therefore, there is a need for a simple system that demonstrates how data can be passed to a program through the command line. The **Argument Passing** project aims to develop a C-based program that accepts command-line arguments, identifies and validates the provided inputs, and processes them according to the required operation. The project demonstrates the use of `argc` and `argv` to receive information from the Linux terminal. The system should handle multiple arguments, display the received values, detect missing or invalid inputs, and provide appropriate output or error messages. This project helps users understand how the Linux

shell communicates with a program during execution and provides practical knowledge of command-line input handling and basic Operating System concepts.

3. Objectives

1. To understand the concept of **argument passing** in Linux and C programming.
2. To accept input directly from the **Linux command line** during program execution.
3. To implement the use of **argc and argv** for receiving command-line arguments.
4. To identify and display the **number and values of arguments** provided by the user.

4. Proposed Methodology

The **Argument Passing** project will be designed and implemented as a simple C-based application in a Linux environment. The system will allow users to provide required inputs directly through the Linux terminal while executing the program. These inputs will be received using the `argc` and `argv` parameters of the `main()` function. The program will first identify the number of arguments provided and then validate whether the required inputs are available and in the correct format. After successful validation, the arguments will be processed according to the required operation, and the appropriate result will be displayed to the user. If the user provides missing or invalid arguments, the program will display a suitable error message and guide the user to provide the correct input. The project will be compiled using the GCC compiler and executed through the Linux terminal. This implementation provides a practical understanding of command-line input, argument handling, validation, and program execution in Linux. It also demonstrates how users can interact with applications efficiently without modifying the program's source code.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Command-Line Arguments	Allows users to provide input directly when executing the program.
<code>argc</code>	Determines the total number of command-line arguments provided.
<code>argv[]</code>	Stores and provides access to the actual command-line argument values.
Process Execution	Demonstrates how a program receives information when it starts execution.
Linux Shell	Accepts the user's command and passes the arguments to the C program.
Input Validation	Checks whether the required arguments are provided and valid.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520090033	V Vishwa Teja	Project design, implementation of argument handling using <code>argc</code> and <code>argv</code>

		argv, and input validation.
2520090058	B Sree Nikhil Reddy	Program testing, error handling, output verification, and debugging.
2520090191	Nalla Rohith	Documentation, project flow/design, GitHub repository management, and final project presentation.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating environment for developing and running the project.
C / C++	Main programming language used to implement the Argument Passing system.
GCC Compiler	Compiles the C source code into an executable program.
Linux Terminal	Used to compile, execute, and provide command-line arguments to the program.
GitHub	Used to store, manage, and share the project repository.

8. Expected Outcome

The **Argument Passing** project is expected to successfully demonstrate how information can be passed to a C program through the Linux command line during program execution. The system will accept one or more arguments from the user and process them using the argc and argv parameters. It will display the number of arguments received, the values of the arguments, and the appropriate result based on the required operation. The program will also validate the provided inputs and display clear error messages when arguments are missing, invalid, or incorrectly formatted. The expected output will be simple, clear, and easy to understand through the Linux terminal. Overall, the project will provide practical knowledge of command-line arguments, input validation, data processing, program execution, and basic Linux Operating System concepts.

9. GitHub Repository

GitHub Repository Link: https://github.com/vishwateja015/KLH_CSIT_2026-2027_batch-14_ARGUMENT-PASSING

Repository Created: Yes

☒ Yes ☐ No

Faculty Name and Signature

Faculty Name: M. Raghupathi

Signature: _____

Remarks: _____

Faculty Use

Project Approved: yes

☒ Yes ☐ No

Date: 01/21/2026

Faculty Signature: [Signature]